

# GeoLean: A Structure-Preserving Transpiler that Ports the Full Euclidean *Elements* of GeoCoq to Lean 4

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## Abstract

We present **GeoLean**, a deterministic, structure-preserving transpiler that translates the *Elements/OriginalProofs* library of GeoCoq—a formalization of Euclid’s *Elements* in the Coq/Rocq proof assistant—into Lean 4. GeoLean reaches **234/234 (100%)** of the target lemmas compiling under a single clean rebuild, with **zero** `sorry/admit/sorryAx` occurrences, no axioms beyond the 16 of the source theory, and a *uniform* proof-search budget applied identically to all proofs. Our central design commitment is *faithfulness without per-lemma manual proof*: the geometric axioms and  $\sim 25$  derived definitions are hand-translated once; a 643-line Lean tactic library re-implements GeoCoq’s `eauto`-backed proof vocabulary (`conclude`, `conclude_def`, `forward_using`) as *bounded*, head-indexed search; and an 860-line Python transpiler performs a line-for-line structural translation of all 234 proofs (which together invoke `conclude` 6,234 times, `conclude_def` 2,002 times, and `forward_using` 3,247 times). No generated proof file is ever hand-edited; every correctness gap is closed by a *general* fix to the transpiler or the tactic library. We give the full engineering taxonomy of the final 16 failures that separated 84.6% from 100% coverage—each a reusable translation rule with before/after evidence—and argue that the combination of a uniform budget and a no-hand-edit invariant is what makes the resulting corpus an *honest*, machine-auditable artifact rather than a collection of bespoke reconstructions.

## 1 Introduction

Cross-prover library translation is a recurring bottleneck in interactive theorem proving: mature formalizations are locked to a single system, and re-deriving them elsewhere is expensive and error-prone. GeoCoq [Boutry et al., 2019] is among the largest formalizations of synthetic geometry in existence—over four thousand lemmas spanning Tarski’s axiomatic geometry and a faithful development of Euclid’s *Elements*. Its *Elements/OriginalProofs* sub-library (234 lemmas in our target set) is a self-contained, axiomatically grounded account of Euclid Book I through the theory of areas, written in a strikingly uniform “assert-and-discharge” proof style.

We ask a sharp question: *can an entire axiomatic-geometry library be ported to another prover deterministically—without a learned model proposing tactics per goal, and without a human writing a single proof by hand—while remaining machine-checkably honest?* We answer affirmatively for *Elements/OriginalProofs*, translating all 234 lemmas to Lean 4.

**Why this is hard.** The naive view—“GeoCoq proofs are flat `assert` chains, so transliterate them”—fails for two reasons. First, the closing tactics (`conclude L` and friends) are not syntactic: they are *searches* against a hint database, and a faithful Lean re-implementation must reproduce their search *semantics* without inheriting Coq’s hint engine. Reproduced naively, these searches detonate Lean’s heartbeat budget on the  $\sim 100$ -hypothesis contexts that deep geometry proofs accumulate. Second, the source uses Coq tactic idioms with no Lean analogue (Coq-internal hypothesis names, `unfold ... in *`, `decompose`, `by cases on`, `n`-ary `splits`), each of which the transpiler must map to a *semantically* equivalent Lean construction, not a token-level copy.

## Contributions.

- **A structure-preserving transpiler** (GeoLean, 860 Python LOC) that maps GeoCoq’s proof vocabulary one-to-one onto a small Lean tactic library, producing proofs that mirror the source line-for-line (Section 3, Listing 1).
- **A bounded port of GeoCoq’s search vocabulary** (643 Lean LOC), in which `conclude/conclude_def/forward_using` become head-indexed, backtracking tactics whose failures are *cheap*, eliminating the timeout cliff (Section 4).
- **A discipline of general fixes.** We never edit a generated proof. The 16 failures that separated 84.6% from 100% are each resolved by one reusable translation rule, which we catalogue with before/after listings as a field guide (Section 5).
- **An honest, auditable artifact and methodology.** A clean-rebuild audit counts only the 234 target oleans, ignoring infrastructure; the final result is 234/234 with no sorry, no extra axioms, and a uniform  $8 \times 10^5$  heartbeat budget (Section 6).

## 2 Background

**GeoCoq and the two stylistic worlds.** A prerequisite measurement shaped this work. GeoCoq is *not* stylistically uniform. Its *Elements/OriginalProofs* proofs are chains of `assert (P)` by (`conclude L`) steps closed by a final `close`; across the 234 target lemmas these invoke `conclude` 6,234 times, `conclude_def` 2,002 times, and `forward_using` 3,247 times. By contrast, the much larger *Main/Tarski\_dev* (1,527 lemmas) is written in idiomatic, goal-directed Coq (`induction/rewrite/destruct`) with *zero* `conclude` uses. The assert-chain style is what makes a deterministic transpiler viable: each `assert` is a self-contained sub-goal whose proof is a named lemma plus context, so the proof *structure* can be copied verbatim and only the closing tactics need a semantics-aware translation. This paper targets the transpilable world; the goal-directed world is future work (Section 8).

**Target axiomatic base.** The source theory is a linear hierarchy of Lean type classes,

$$\text{neutral\_basis} \subset \text{neutral} \subset \text{ruler\_compass} \subset \text{euclidean} \subset \text{area},$$

comprising the 16 geometric axioms of the development—including Euclid’s fifth postulate (`postulate_Euclid5`, a field of `euclidean`) and the three area-pasting axioms (`axiom_paste3/4/5`, fields of `area`). We hand-translate these axioms and the  $\sim 25$  derived definitions they need (`Col`, `BetS`, `Par`, `CongA`, `Meet`, `OS`, `TS`, `PG`, `EF`, ...) once into Lean; everything downstream is generated.

Table 1: Core statement-translation rules. “ $L$ ” is a lemma name, “ $X$ ” a definition.

| GeoCoq statement                                  | Lean emission                                |
|---------------------------------------------------|----------------------------------------------|
| <code>assert (T) by (conclude L)</code>           | <code>have : T := by conclude L</code>       |
| <code>assert (∃...) by (...); destruct ...</code> | <code>obtain ⟨ws, _⟩ : ∃... := by ...</code> |
| <code>by cases on (D) + {...} blocks</code>       | <code>rcases (show D by ...) with ...</code> |
| <code>assert (T). {...} (T a negation)</code>     | <code>have : T := by intro h; ...</code>     |
| <code>assert (T). { eapply L. exact ... }</code>  | <code>have : T := by conclude L</code>       |
| <code>close / contradict</code>                   | <code>close / contradict</code>              |

Listing 1: lemma\_3\_6a (Coq, left) and its GeoLean translation (Lean, right). The proof structure is copied verbatim; only the closing tactics carry semantics.

```
(* GeoCoq *)
Lemma lemma_3_6a :
  forall A B C D,
    BetS A B C -> BetS A C D ->
    BetS B C D.
Proof. intros.
  assert (BetS C B A) by
    (conclude axiom_betweennesssymmetry).
  assert (BetS D C A) by
    (conclude axiom_betweennesssymmetry).
  assert (BetS D C B) by
    (conclude axiom_innertransitivity).
  assert (BetS B C D) by
    (conclude axiom_betweennesssymmetry).
close. Qed.

-- GeoLean output (Lean 4)
theorem lemma_3_6a :
  (A B C D : Point),
  BetS A B C → BetS A C D →
  BetS B C D := by
  intro A B C D h1 h2
  have : BetS C B A := by
    conclude axiom_betweennesssymmetry
  have : BetS D C A := by
    conclude axiom_betweennesssymmetry
  have : BetS D C B := by
    conclude axiom_innertransitivity
  have : BetS B C D := by
    conclude axiom_betweennesssymmetry
close
```

### 3 The GeoLean transpiler

**Pipeline.** For each lemma, GeoLean (i) parses the Coq statement into a Lean signature and an `intro` prelude by splitting the leading `forall` binders and `->` hypotheses; (ii) detects the source axiom class (from `Context` ‘{Ax: ...}’) and emits the matching `open/variable` preamble; (iii) splits the proof body into statements at brace/period boundaries at paren-depth zero; and (iv) translates each statement by syntactic class. Table 1 lists the core statement-level rules.

**Worked example.** Listing 1 shows a complete lemma and its generated translation; the correspondence is line-for-line, which is precisely what makes faithfulness auditable by inspection rather than by trust.

### 4 The tactic library: GeoCoq search, made bounded

The semantic core of the port is the closing-tactic library. In GeoCoq, `conclude L` means “apply lemma  $L$  and discharge its premises from the context,” realized as an `eauto` call against a curated hint database; `conclude_def X` unfolds a definition  $X$  and discharges by the same search; `forward_using L` applies  $L$  whose conclusion is a conjunction and selects the matching conjunct. A direct Lean rendering as `aesop/solve_by_elim` over the full context is correct but *ruinously slow*: a single call triggers seconds of typeclass and `whnf` search on the  $\sim 100$ -hypothesis contexts of deep proofs, and the cumulative cost blows the heartbeat budget.

Our `conclude` (Listing 2) instead normalizes the context once and then walks an *ordered*

Listing 2: The `conclude` macro (abridged). The preamble normalizes the context; the cascade tries the cheapest closers first and reaches the bounded, head-indexed `conclude_bounded` (the faithful `eauto` port) before any unbounded search.

```
macro "conclude" t:term : tactic =>
  '(tactic|
    ((try splitter); (try remove_double_neg); (try expose_ncol);
      first
        | done | assumption
        | (subst_vars; (first | assumption | rfl)) -- equality idioms
        | exact $t
        | (apply $t <;> assumption)
        | (apply $t <;> assum_inj) -- disjunctive premises (area axioms)
        | conclude_bounded $t -- head-indexed bounded eauto
        | forward_bounded $t -- conjunct projection, bounded
        | solve_by_elim [$t, And.left, And.right]
        | (apply $t <;> (first | assumption | aesop)) -- last resort
        | aesop))
```

*cascade of cheap-failing alternatives*, ending in a *bounded* head-indexed search that is the faithful analogue of `eauto`: it applies  $L$  and discharges each premise by backtracking over only those hypotheses whose head symbol matches the premise (so a `BetS` premise consults only `BetS` hypotheses). A failure of the bounded search is linear in the matching hypotheses, never an unbounded blow-up. The unbounded `aesop` survives only as a last resort, itself capped by its own rule-application bound.

The library further provides: `splitter` (split conjunction *hypotheses* so bundled facts become directly addressable); `splits` (an *elaborator* that splits only syntactic *And goals*, leaving definitional predicates such as `nCol` atomic); `expose_ncol/Col_nCol_False` (cheap collinearity closers that fire *before* any search); and `assum_inj` (Section 5, item 8).

## 5 From 84.6% to 100%: a taxonomy of general fixes

The transpiler reached 84.6% (198/234) on first assembly. The remaining 16 failures were *not* resolved by hand-writing proofs; each exposed a missing *general* translation rule. We give the eight rule classes with concrete before/after evidence, since they constitute a reusable guide for porting other assert-chain corpora.

**(1) Nested-paren capture.** The payload extractor for `by (...)` was non-greedy and truncated at the first inner parenthesis, so `apply (L args);assumption` lost its tail. *Fix*: balanced (greedy) capture.

**(2) Focus-block polarity & eapply-blocks.** A `{...}` block proving a *positive* goal was uniformly mis-typed as a negation and prefixed with `intro h`; moreover, blocks of the form `simple eapply L. exact H40. ...` reference Coq-*internal* hypothesis names (H40) that do not exist in Lean. *Fix*: emit `intro h` only for negation types, and recognize an `eapply-then-exact` block as exactly `conclude L`:

```
-- before (wrong) -- after (general)
have : Par A b d C := by have : Par A b d C := by
  intro h -- bogus (Par is not a Pi) conclude proposition_30A
  apply proposition_30A -- exact H40 dropped
```

**(3) Explicit-argument apply.** A deliberate `apply (L a b c);assumption`—the author pinned arguments to avoid search—was over-eagerly collapsed to `conclude L`, which then re-derived the arguments by unification over a  $\sim 160$ -hypothesis context and timed out. *Fix:* preserve the pinned arguments; fall back to `conclude` only when direct application cannot unify (e.g.  $L$  concludes a conjunction and the goal is one conjunct):

```
have : Par A b C d := by
  (first | (apply proposition_30A _ _ _ E f G H K <;> assumption) | conclude
    proposition_30A)
```

**(4) De Morgan pre-emission.** In a  $\neg\neg(D_1 \vee \dots \vee D_n)$  block, the body needs  $\neg D_i$ , which `conclude` otherwise synthesizes as  $h \circ \text{Or.in}^x$  under search over the full context—an  $8 \times 10^5$ -heartbeat timeout. *Fix:* emit each leaf negation *eagerly* after `intro h`, turning a search into an  $O(1)$  assumption:

```
have :  $\neg \neg$  (CR A f G H CR A E G H) := by
  intro h
  have :  $\neg$  (CR A f G H) := fun c h (Or.inl c)      -- pre-emitted
  have :  $\neg$  (CR A E G H) := fun c h (Or.inr c)      -- pre-emitted
  have : CR A E G H := by conclude lemma_30helper   -- now matches directly
  contradict
```

**(5) Definition elimination.** Three Coq idioms—`auto using parnotmeet`, `unfold X in H; decompose [ex and] H; auto, and [remove_double_neg,] unfold X in *; assumption`—all mean “extract a conjunct from definition  $X$ .” (Indeed the GeoCoq helper `parnotmeet` is itself proved by `conclude_def Par`.) *Fix:* map each to `conclude_def X`, which performs the unfold, any double-negation stripping, and the conjunct extraction in one bounded step.

**(6) Hierarchy scoping.** An area-class proof using `postulate_Euclid5` (an euclidean-class field) failed name resolution because the intermediate class was not `opened`. *Fix:* open the *entire* class chain up to the detected class (`...ruler_compass euclidean area`), not just the leaf.

**(7) Apply-family normalization.** `unshelve epose (L ...)`, `eapply (L)` with ... (bare or parenthesized instantiations), and leading-parenthesis lemma heads all reduce to `conclude L`. *Fix:* a robust apply-head extractor that strips `@`, parentheses, `with`-clauses, and trailing closers before identifying the lemma.

**(8) Disjunctive premises (area axioms).** `axiom_paste3/4/5` carry premises such as `BetS A M B  $\vee$  A = M  $\vee$  M = B`, whose midpoint  $M$  is left *unpinned* as a metavariable after `apply`, while the pinning fact `BetS B E C` sits *bundled* inside a conjunction hypothesis. *Fix:* the preamble `splitter` unbundles the determining ground `BetS` premise (pinning  $M$  from an earlier, ground subgoal); a small `assum_inj` closer then discharges the now-concrete branch by `assumption/rfl`—branch injection, never search:

```
macro "assum_inj" : tactic => '(tactic|
  first | assumption | rfl
    | (left; (first | assumption | rfl))
    | (right; (first | assumption | rfl
      | (left; (first | assumption | rfl))
      | (right; (first | assumption | rfl))))))
```

Every item is a *rule over a syntactic class*, validated by re-running the transpiler and performing a full clean rebuild; none is keyed to a specific lemma. This is the methodological claim of the paper: structure-preserving porting of an assert-chain corpus reduces to a finite, discoverable set of such rules.

Table 2: Coverage progression of *Elements/OriginalProofs* (234 lemmas). Each row is an independent clean-rebuild audit (delete all target oleans, rebuild, count). “Own” = own-error count; “Casc.” = cascade count.

| Milestone                            | Compiled   | %            | Own      | Casc.    | Fix class added                 |
|--------------------------------------|------------|--------------|----------|----------|---------------------------------|
| Baseline assembly                    | 198        | 84.6         | 4        | 32       | —                               |
| bundled-conjunction <b>close</b>     | 199        | 85.0         | 4        | 31       | <b>splitter</b> in <b>close</b> |
| block polarity / nested apply        | 215        | 91.9         | 4        | 3        | taxonomy 1–2                    |
| de Morgan / def-elim / explicit-args | 223        | 95.3         | 4        | 7        | taxonomy 3–5                    |
| hierarchy / apply-normalization      | 230        | 98.3         | 1        | 3        | taxonomy 6–7                    |
| disjunctive premises                 | <b>234</b> | <b>100.0</b> | <b>0</b> | <b>0</b> | taxonomy 8                      |

## 6 Evaluation

**Coverage progression.** Table 2 traces the path from 84.6% to 100%. The decisive structural property is the *cascade*: the audit distinguishes an *own-error* (a lemma whose own proof fails, with all dependencies compiled) from a *cascade* (a lemma blocked transitively through the import DAG by a broken dependency). Because a single own-error gates an entire downstream cluster, each general fix releases many lemmas at once—most dramatically, repairing the two parallel-line lemmas **proposition\_30/35A** unblocked the whole prop 35–48 area/Pythagoras family (a jump of +8), and the final area-axiom fix closed the last own-error and its three cascades together.

**Honesty audit and methodology.** We treat “it compiles” as insufficient and audit the artifact against four explicit criteria. (i) *Target-only coverage*. The audit script deletes every target olean, performs one clean **lake build** of the aggregate module, and counts only the 234 lemma oleans—*never* the infrastructure modules (**euclidean\_tactics**, **euclidean\_defs**, **conclude\_bounded**). This guards against inflating coverage with helper modules; the build system’s raw “Built” count is explicitly *not* our metric. (ii) *No escape hatches*. A scan of all generated proofs finds **0** occurrences of **sorry/admit/sorryAx**. (iii) *Uniform budget*. Every proof carries the *same* **maxHeartbeats** =  $8 \times 10^5$  ( $4 \times$  Lean’s default), never bumped per-lemma to mask a fixable tactic; under our discipline a timeout is a transpiler bug to be fixed *generally*, not hidden locally. (iv) *Axiom hygiene*. No proof depends on any axiom beyond the 16 hand-translated geometric axioms of the source theory. We regard criteria (iii) and the no-hand-edit invariant as the two pillars of the honesty claim.

**Cost of the discipline.** Refusing per-lemma edits is not free: a single missing rule can leave several lemmas red, and each fix requires re-running the transpiler over the affected files plus a (sometimes full) clean rebuild—infrastructure changes such as the **assum\_inj** addition force a recompile of all 234 lemmas. We accept this cost because it is exactly what guarantees the rules *generalize*: a fix that survives a clean rebuild of the whole corpus cannot be an overfit patch.

## 7 Related work

Library translation has been studied via shallow embeddings, proof-term replay, and foundational alignment (e.g. between HOL systems, and from HOL to Coq/Isabelle). Learned autoformalization and neural premise selection translate *statements* or propose tactics per goal. GeoLean differs on two axes: it is *deterministic* (no per-goal model) and *structure-preserving* (it copies proof structure rather than re-searching for a proof), which is feasible

precisely because the source library is written in a uniform assert-chain discipline. The closest relatives are tactic-language transliterators; our contribution is to show that, for a real axiomatic-geometry corpus, the residual semantic gap is a small, finite set of rewrite rules over the closing-tactic vocabulary, and that closing that gap to 100% is dominated by reproducing `eauto`’s search behaviour *within* a heartbeat budget.

## 8 Limitations and future work

GeoLean’s reach is bounded by the assert-chain style. A spike on *Main/Tarski\_dev* (1,527 lemmas) found that  $\sim 90\%$  of those proofs are goal-directed and *not* structure-copyable; that corpus needs a reasoning agent (an LLM-driven, dependency-ordered pipeline), with the transpiler reserved for the  $\sim 3\%$  that are pure assert-chains. Our uniform heartbeat budget, while honest, is generous; tightening it would surface further opportunities to strengthen the bounded tactics and shrink the `aesop` tail. Finally, statement faithfulness is currently checked by a definitional-equality harness and by inspection of the line-for-line correspondence; a machine-checked round-trip (Lean statement  $\rightarrow$  Coq) would further harden the faithfulness guarantee.

## 9 Conclusion

A real axiomatic-geometry library—the full Euclidean *Elements* of GeoCoq, 234 lemmas—can be ported to Lean 4 *deterministically* and *honestly*: no learned model in the loop, no hand-written proofs, no `sorry`, no per-lemma escape hatches, a single uniform search budget. The enabling insight is that a uniform, search-backed proof style turns translation into a finite discovery problem over a small tactic vocabulary, in which each remaining failure yields a *general* rule and each rule, surviving a full clean rebuild, provably generalizes. We hope the rule taxonomy of Section 5 and the bounded-search design of Section 4 serve as a practical blueprint for structure-preserving porting of other assert-chain corpora.

## References

- P. Boutry, G. Braun, J. Narboux, et al. Formalization of the arithmetization of Euclidean plane geometry and applications. *Journal of Symbolic Computation*, 90:149–168, 2019.